



HEXAGON

**Actran for Exterior
Aeroacoustic Problems**

BANC Tandem Cylinders

STAR-CCM+/Actran for Computational Aero-Acoustics”, at Star American Conference 2011

Introduction

- The objective is to solve an academic test case, representative of a simplified landing gear (tandem cylinders)
- The CFD is very challenging in such an application
- Many previous works were based on FWH tools
- We believe that the use of FWH tools adding difficulty to the CFD, and we present an alternative method → Actran AeroAcoustics
- The presentation shows
 - A successful comparison between FWH solution and Actran solution
 - A second Actran computation with a much coarser CFD input

Tandem cylinder benchmark

- **BANC**: Benchmark problems for Airframe Noise Computations. A workshop is organized yearly by Technical Committees of AIAA, jointly with the AIAA-CEAS Aeroacoustics Conference
- The BANC Tandem Cylinders model is available here:
https://info.aiaa.org/tac/ASG/FDTC/DG/BECAN_files/Workshop_June_2010_Final_Problem_Statements
- The following work is published in the paper “ STAR-CCM+/Actran for Computational Aero-Acoustics”, at Star American Conference 2011

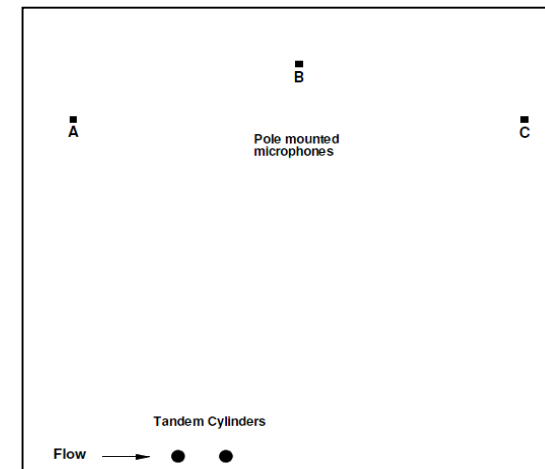
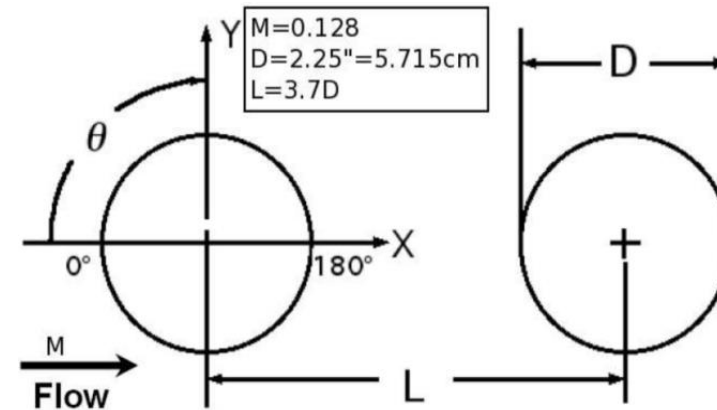
Problem Statement

- Flow condition:
 - Mach: 0.128
 - $U_0 = 44 \text{ m/s} \rightarrow 158 \text{ km/h}$
 - $Re = 1.66 \times 10^5$
- Experimental Configurations:
 - 12D span for flow measurements (BART)
 - 16D span for acoustic measurements (QFF)



Experimental configuration (ref 2.)

Tandem Cylinders - Geometry (ref 1.)

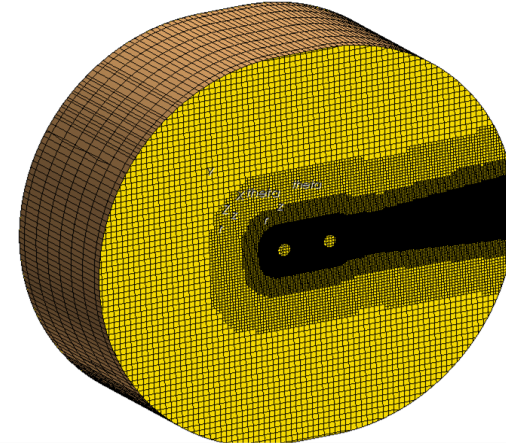


Schematic of microphone locations (ref 2.)

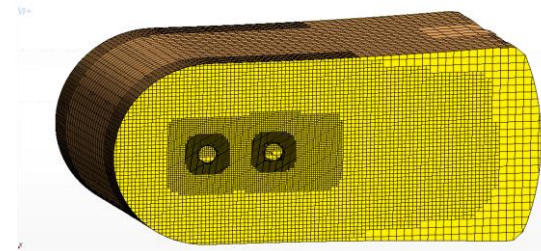
CFD Description

- STAR-CCM+ v5.06.007
- Model sizes
 - CFD model 1: 6.7 million cells (top)
 - CFD model 2: 1.1 million cells (bottom)
- CFD model 2, compared with CFD model 1
 - Has reduced CFD domain + coarser mesh
 - Does not calculate the acoustic field
 - Is only used to compute the (unsteady) flow in the source area
- Obviously CFD model 2 runs much faster
- ... and it could certainly be further improved...

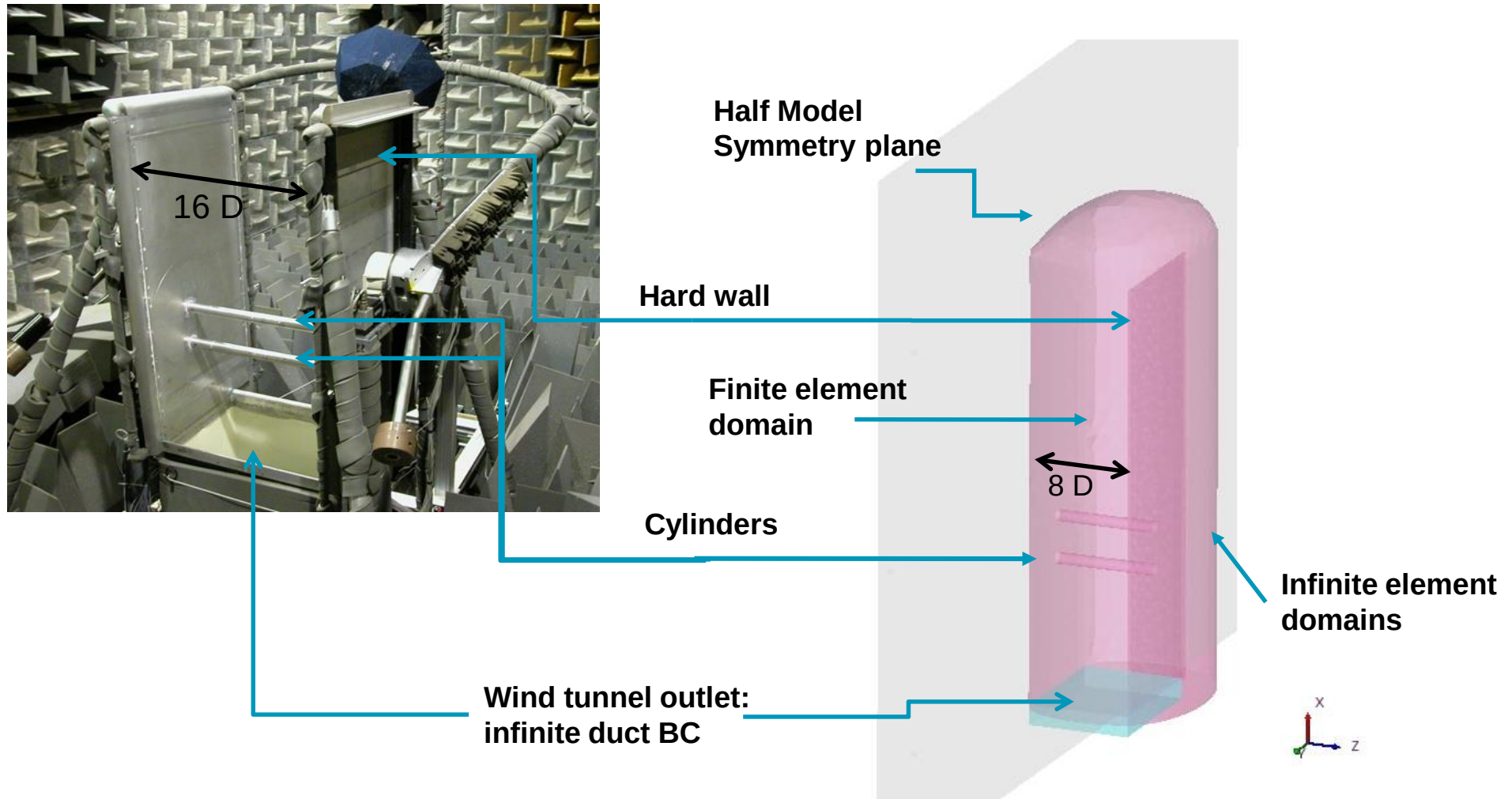
CFD model 1



CFD model 2

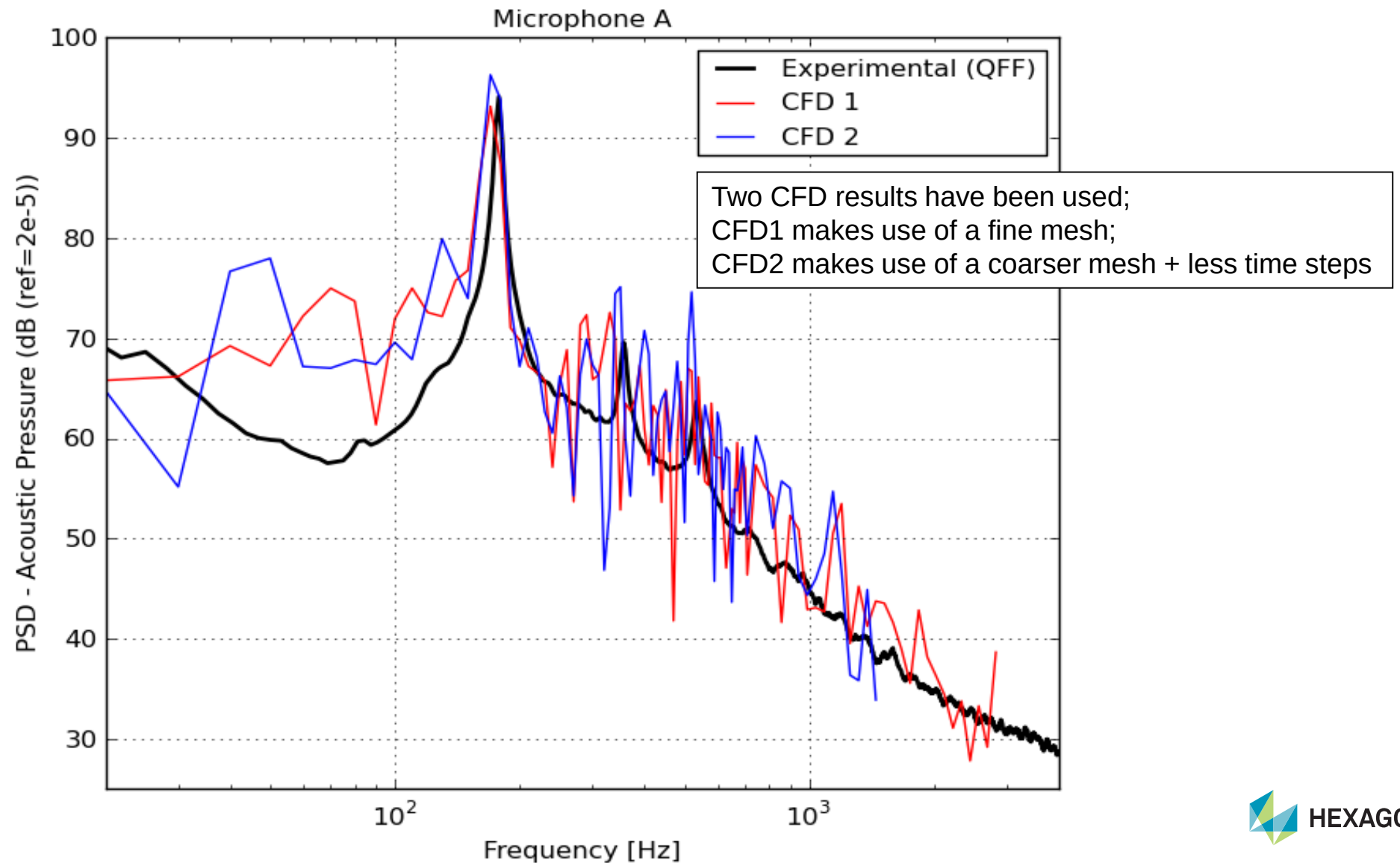


Actran Simulations



Actran setup fully reproduces experimental QFF conditions

Results of Actran Results with CFD from Star-CCM+



Conclusions, Perspectives

- STAR-CCM+ / Actran software combination provides good correlation with experimental results
- Advantages on the CFD
 - No need to refine the CFD grid to propagate acoustic waves → reduced CFD effort
 - Incompressible CFD is enough
- Other advantages
 - Several acoustic investigations can be performed using the same CFD data
 - The acoustic boundary conditions (walls, free field, free duct modes, impedance, etc.) can be imposed independently from the CFD boundary conditions
- Paper published: "STAR-CCM+/Actran for Computational Aero-Acoustics", at Star American Conference 2011

Rod-Airfoil Validation by Daimler

Actran Users Meeting, Brussels, 2012

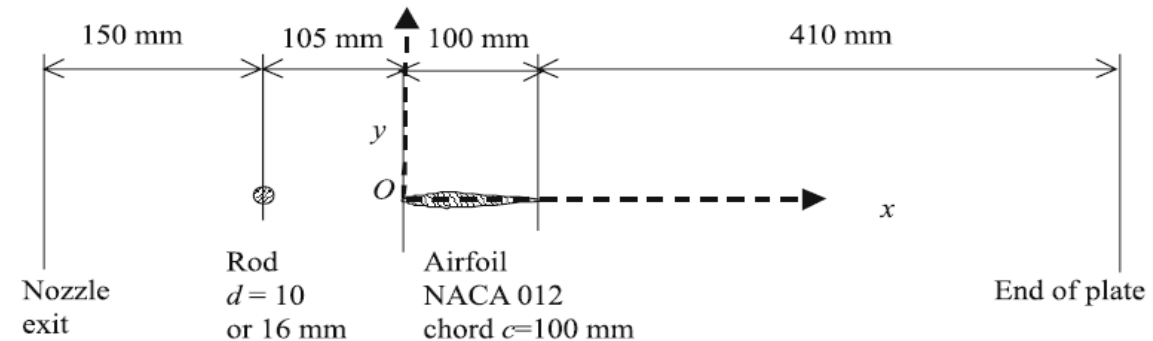
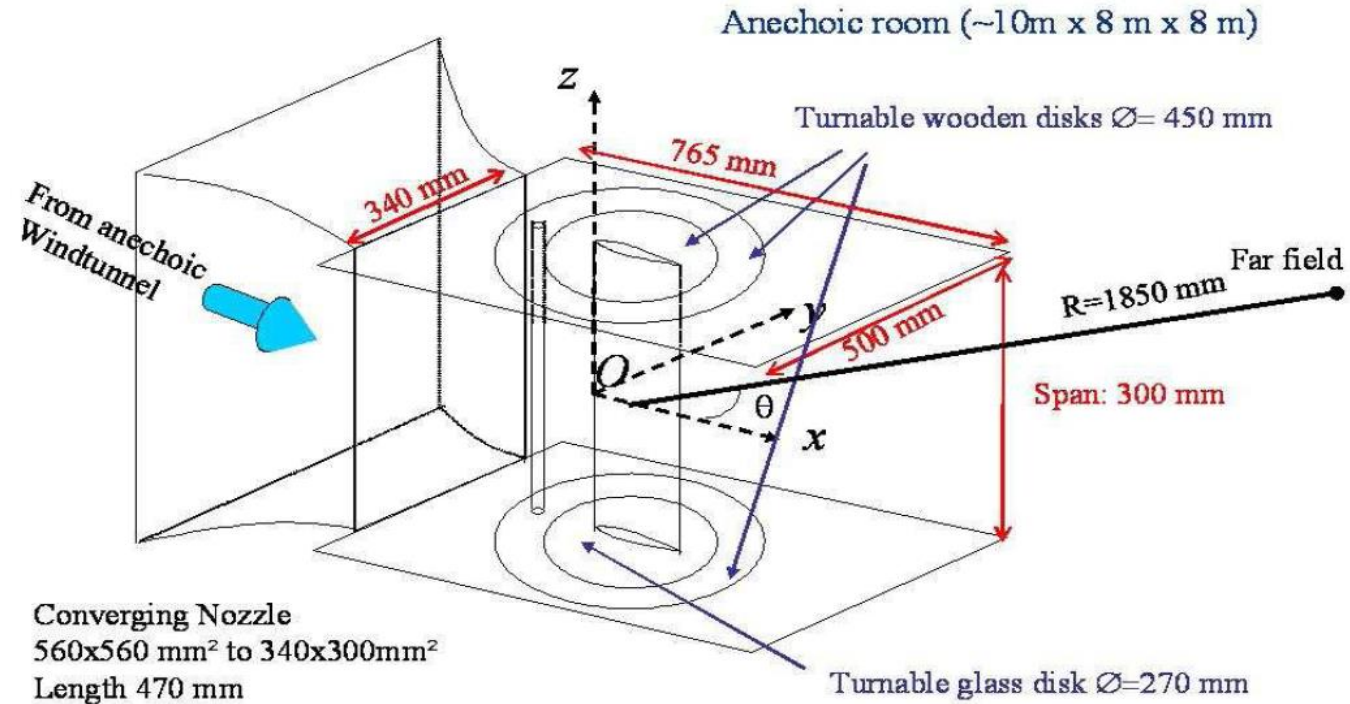
Experiment: Rod-Airfoil

Experimental setup

- Cylinder: $\varnothing = 10 \text{ mm}$
- Airfoil: NACA 0012,
 $c = 100 \text{ mm}$
- Span: 300 mm
- Far field measurements:
 $R = 1.85 \text{ m}$

Experimental data

- Geometric and hydrodynamic boundary conditions
- Velocity profiles and spectra
- Wall pressure and far field spectra



Simulation Setup

DES simulation with Star-CCM+ (V6.05)

- Turbulence model: IDDES Menter SST
- Temporal : 2nd order / Convection: Hybrid-BCD

Meshing

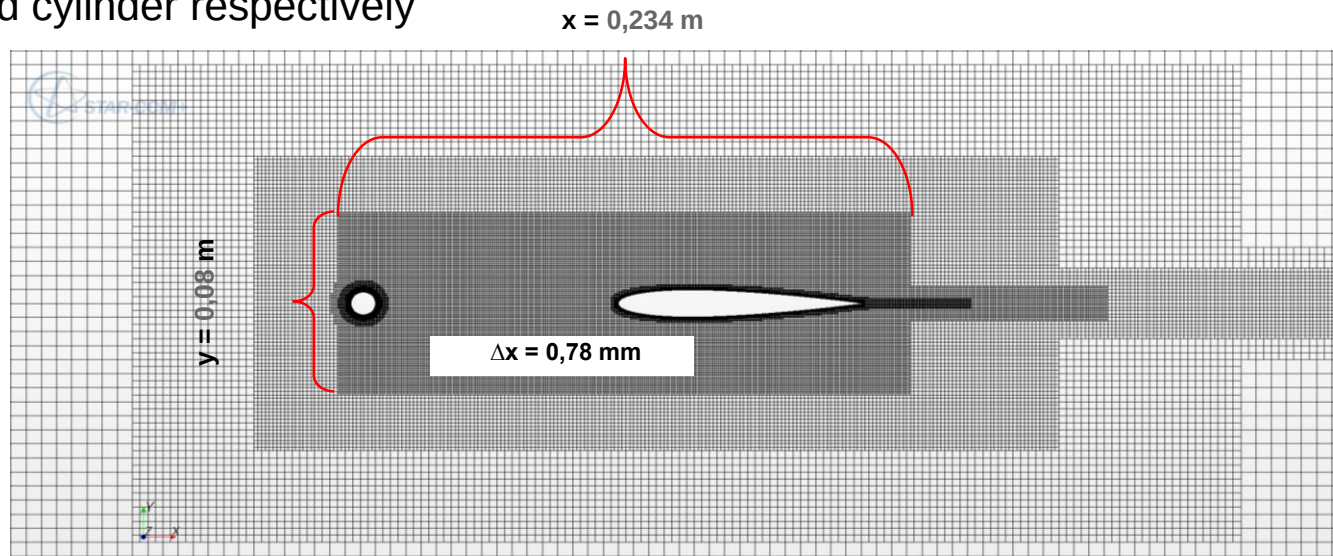
- Computational domain: 6 m x 4 m x 0.3 m (x, y, z)
- Step by step refinement until 0.78 mm in the finest region
- Wall resolution $y^+ < 1$ and 3 at the airfoil and cylinder respectively
- ~ 95 million cells

Inlet boundary conditions

- $T = 293$ K
- $P = 899$ mbar
- $Ma = 0.2098 \rightarrow Re = 44\,000$ (Cylinder)
- Turbulent intensity = 0.78 %

Simulation time

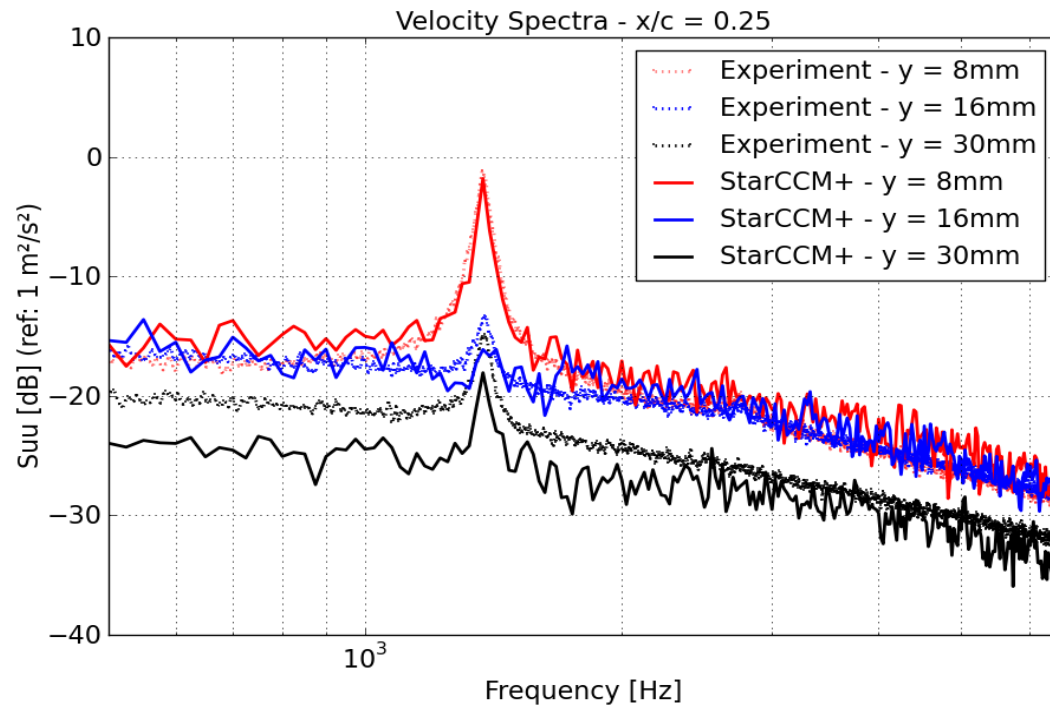
- $T = 0.2$ s with $\Delta t = 1 \cdot 10^{-5}$ s



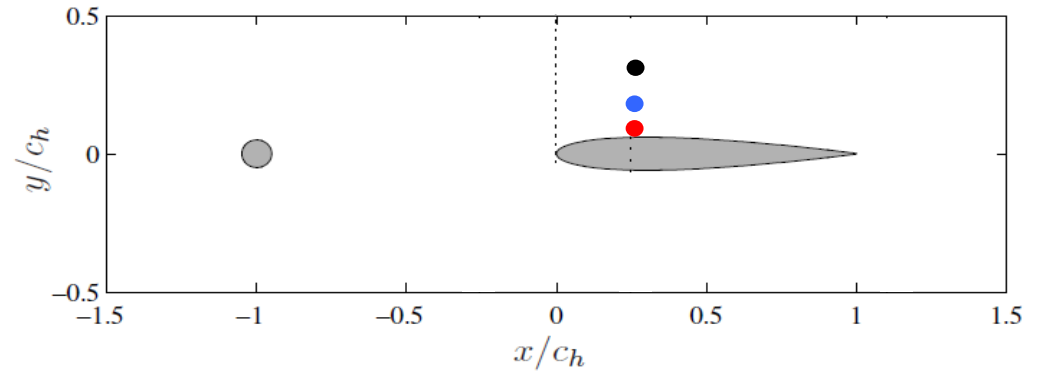
Mesh resolution

Fluid Dynamics

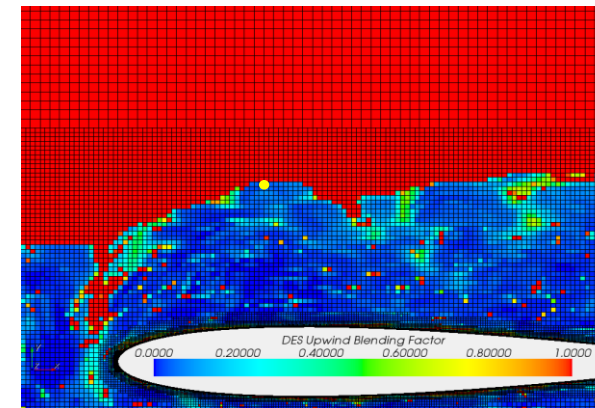
Velocity spectra at $x/c = 0.25$



- Good agreement with experimental data
- Slightly lower level at $y = 30 \text{ mm}$ caused by the transition between LES and RANS at this distance

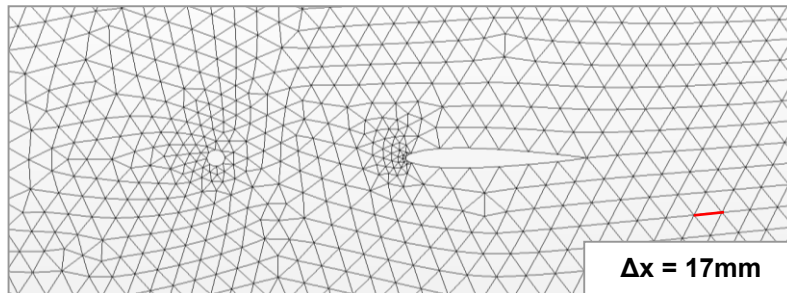
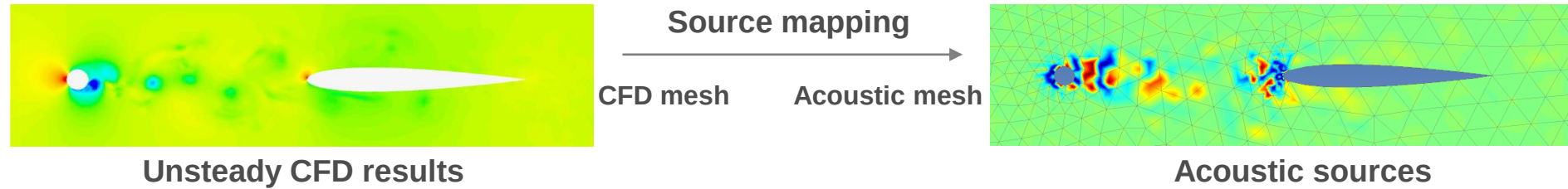


$x = 25\text{mm} / y = 30\text{mm}$

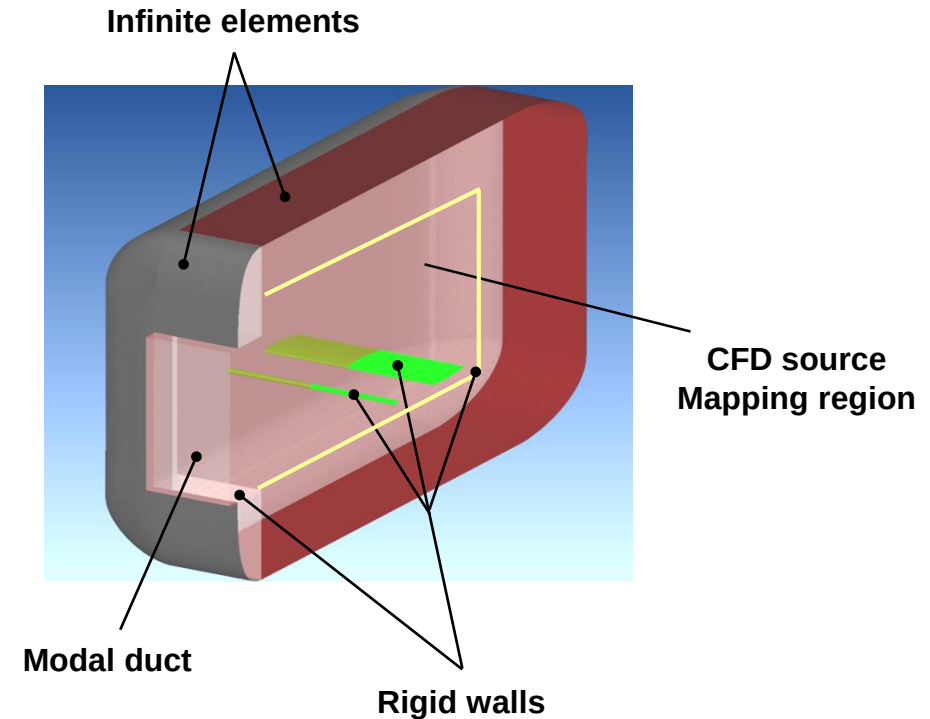


DES Upwind Blending Factor

Hybrid CAA



Acoustic mesh only need to represent the geometry correctly.
It does not need to be refined at source intensive region



Comparison with Experiment

